```
dims@ubuntu-VM:~/iac-homework$ sudo docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED
          STATUS          PORTS
NAMES
53b5150e6806   nginx:latest          "/docker-entrypoint.…"   2 minut
es ago    Up 2 minutes    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
nginx-lb
fe21d2a52b45   custom-web-app:1.0    "python app.py"          14 minu
tes ago   Up 14 minutes   5000/tcp
web-app
d24e66445866   postgres:15           "docker-entrypoint.s…"   17 minu
tes ago   Up 17 minutes   5432/tcp
postgres-db
dims@ubuntu-VM:~/iac-homework$
```

```
dims@ubuntu-VM:~/iac-homework$ sudo docker exec -it nginx-lb curl ht
tp://web-app:5000
Web app works. Connected to PostgreSQL: PostgreSQL 15.17 (Debian 15.
17-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.
0-19) 14.2.0, 64-bit
dims@ubuntu-VM:~/iac-homework$
```

```
dims@ubuntu-VM:~/iac-homework$ curl http://localhost:8080
Web app works. Connected to PostgreSQL: PostgreSQL 15.17 (Debian 15.
17-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.
0-19) 14.2.0, 64-bit
dims@ubuntu-VM:~/iac-homework$
```

```
dims@ubuntu-VM:~/iac-homework/terraform$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding kreuzwerker/docker versions matching "~> 3.0"...
- Installing kreuzwerker/docker v3.9.0...
- Installed kreuzwerker/docker v3.9.0 (self-signed, key ID 0DCE69892
7DAF8EC)
Partner and community providers are signed by their developers.
If you'd like to know more about provider signing, you can read abou
t it here:
https://developer.hashicorp.com/terraform/cli/plugins/signing
Terraform has created a lock file .terraform.lock.hcl to record the
provider
selections it made above. Include this file in your version control
repository
so that Terraform can guarantee to make the same selections by defau
lt when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform pla
n" to see
any changes that are required for your infrastructure. All Terraform
 commands
should now work.

If you ever set or change modules or backend configuration for Terra
form,
rerun this command to reinitialize your working directory. If you fo
rget, other
commands will detect it and remind you to do so if necessary.
```

```
dims@ubuntu-VM:~/iac-homework/terraform$ sudo docker inspect --forma
t='{{.ID}}' nginx-lb
a54bc2c233bd5d401058ba70a292fe7b0f82073afe54bb209e39fca8b5c58510
dims@ubuntu-VM:~/iac-homework/terraform$ sudo docker inspect --forma
t='{{.ID}}' nginx-lb
a54bc2c233bd5d401058ba70a292fe7b0f82073afe54bb209e39fca8b5c58510
dims@ubuntu-VM:~/iac-homework/terraform$ sudo docker inspect --forma
t='{{.ID}}' postgres-db
515f213e8450e9dd19a13c0ba1a0b3adfb5045b4039b90b85799b36e1f1b3e61
dims@ubuntu-VM:~/iac-homework/terraform$
```

```
dims@ubuntu-VM:~/iac-homework/terraform$ terraform plan

Terraform used the selected providers to generate the following
execution plan. Resource actions are indicated with the following
symbols:
  + create

Terraform will perform the following actions:

  # docker_container.nginx will be created
  + resource "docker_container" "nginx" {
      + attach                                      = false
      + bridge                                      = (known after a
pply)
      + command                                     = (known after a
pply)
      + container_logs                              = (known after a
pply)
      + container_read_refresh_timeout_milliseconds = 15000
      + entrypoint                                  = (known after a
pply)
      + env                                         = (known after a
pply)
      + exit_code                                   = (known after a
pply)
      + hostname                                    = (known after a
pply)
      + id                                          = (known after a
pply)
      + image                                       = "nginx:latest"
      + init                                        = (known after a
pply)
      + ipc_mode                                    = (known after a
pply)
      + log_driver                                  = (known after a
```

```
      + healthcheck (known after apply)

      + labels (known after apply)

      + networks_advanced {
          + aliases      = []
          + name         = "app-network"
            # (3 unchanged attributes hidden)
        }
    }

  # docker_network.app_network will be created
  + resource "docker_network" "app_network" {
      + driver      = (known after apply)
      + id          = (known after apply)
      + internal    = (known after apply)
      + ipam_driver = "default"
      + name        = "app-network"
      + options     = (known after apply)
      + scope       = (known after apply)

      + ipam_config (known after apply)
    }

  # docker_volume.postgres_data will be created
  + resource "docker_volume" "postgres_data" {
      + driver     = (known after apply)
      + id         = (known after apply)
      + mountpoint = (known after apply)
      + name       = "postgres-data"
    }

Plan: 5 to add, 0 to change, 0 to destroy.
```

```
4 directories, 8 files
dims@ubuntu-VM:~/iac-homework$ tree
.
├── app
│   ├── app.py
│   ├── Dockerfile
│   └── requirements.txt
├── nginx
│   └── default.conf
└── terraform
    ├── main.tf
    ├── outputs.tf
    ├── terraform.tf
    └── variables.tf

4 directories, 8 files
dims@ubuntu-VM:~/iac-homework$
```